

Taehyun Park

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GitHub: github.com/iamtaehyunpark

RESEARCH INTERESTS

AI Agent Dynamics, LLM Architectures, Agentic Memory & Knowledge Representation, Retrieval-Augmented Generation (RAG), Uncertainty Quantification.

EDUCATION

University of Wisconsin - Madison

B.S. in Computer Science

Cumulative GPA: 3.96/4.00

Honors: Dean's List (All Semesters), Honors in Major (Computer Science)

Madison, WI

Expected May 2027

RESEARCH EXPERIENCE

Failure Attribution in Multi-Agent LLM Systems, Yonsei University

Undergraduate Research Intern

Seoul, South Korea

Jun 2026 – Current

- Extended single-agent uncertainty estimation to multi-agent LLM systems, reframing failure attribution as localization over a calibrated, typed per-step error field rather than text-based verdict extraction.
- Disentangled semantic failures from systemic failures to reduce noise in uncertainty quantification during multi-agent failure localization.
- Benchmarked against published verdict-based and trained-tracer baselines on the Who&When dataset under a corrected scoring metric, and curated a trajectory dataset spanning diverse multi-agent frameworks.

Uncertainty Quantification for Single-Agent LLM Systems

Undergraduate Researcher

Madison, WI

Jun 2026 – Aug 2026

- Developed an amortized methodology for step-level uncertainty quantification in single-agent LLM systems: characterizing one measurement instrument once and deploying it across arbitrary target agents.
- Ran ~30k labeled agent steps across 6 model families and 2 benchmarks, establishing that no intrinsic UQ signal transfers reliably across models or tasks and that trajectory-level evaluation is confounded by episode length.
- Selected an external LLM as the characterized instrument via an empirical instrument-selection study, then validated its recipe invariance and transfer through sequential gates: mid-size model score-channel vs. frontier llm-as-a-judge comparison, and label-free operating-point transfer.
- Designed a two-phase online deployment mechanism (pre-execution quantile gating with retrospective calibration).

EngramTrace — Non-Parametric Long-Term Memory for LLM Agents

Sole Author

Madison, WI

Feb 2026 – May 2026

- Conceived a novel answer to the LLM memory problem: instead of vector stores or graph databases, the agent's entire knowledge lives in a single mutable HTML document, a substrate an LLM can read, restructure, and cross-link as a whole, turning knowledge maintenance from thousands of per-edge writes into a handful of batch operations per consolidation cycle.
- Designed the system around a neuroscience-inspired memory loop(encoding, retrieval, and reconsolidation) so knowledge accumulates continuously from live interaction: conversations are buffered, topic shifts detected via embedding drift, and new facts surgically merged into the persistent structure with no retraining and no re-indexing.
- Invented Hierarchical Embedding Summation, where each knowledge node's vector inherits geometri-

cally decaying context from its ancestors, so retrieval understands where a fact sits in the knowledge tree, resolving the semantic-isolation failure of flat RAG.

- Demonstrated the architecture end-to-end on a demanding multi-hop QA benchmark by streaming every passage through the live consolidation pipeline validated on the full evaluation split with the official scorer, entirely on self-hosted open models.

Last-Mile Delivery Living Lab, Yonsei University

Research Assistant / Participant

Seoul, South Korea

Nov 2022 – Jan 2023

- Investigated the integration of autonomous delivery robots into real-world social infrastructure and university testbed environments.
- Participated in live technical demonstrations of delivery robots and systemically analyzed driving failures and edge-case obstacle recognition issues.
- Reprogrammed on-board sensors and Arduino microcontrollers based on hardware failure logs to optimize local path-planning and reactive obstacle avoidance.
- Successfully integrated sensor updates back into production models, demonstrably reducing routing and localization error rates.

PROJECTS & ENGINEERING EXPERIENCE

Issue-In | Independent Project

Apr 2025 – Jul 2025

- Architected an AI-integrated trend platform using Next.js to synthesize real-time data across entertainment, economics, and current affairs into intuitive context-rich dashboards.
- Engineered a proprietary scraping engine and custom ranking algorithm to aggregate and quantify engagement metrics from major online communities in real-time.
- Integrated LLM-driven synthesis to automate analyzing scraped information and generation of background context for trending topics, bridging the information gap found in traditional media.
- Optimized high-concurrency data handling by implementing robust database management practices to process large-scale keyword datasets with low latency.

MILITARY SERVICE & ENGINEERING EXPERIENCE

Republic of Korea Army

C4I System (Server) Management Specialist, Sergeant

Paju, South Korea

Sep 2023 – Mar 2025

- Administered ROKA tactical IT infrastructures (C4I systems), managing high-availability satellite, radio, and optical fiber communications.
- Maintained critical Linux and Windows server nodes for military network defense; configured Unified Threat Management (UTM) and Network Access Control (NAC) policies.
- Authored automated scripting tools and engineered an internal intranet web portal, significantly reducing manual administration latency.
- Completed intensive training at the ROKA Signal School, mastering foundational networking, optical routing hardware, and tactical security.

HONORS, COMPETITIONS & CERTIFICATES

Youth Start-up Competition (Korea SMEs and Startups Agency)

Jul 2024 – Oct 2024

- Designed a wireless communication network system using AI and drones, leveraging network theories and hardware principles.
- Proposed AI application in radio wave processing technology and studied mesh network technology during business planning.
- Completed a development course in Keras AI to implement a proof-of-concept simulation using reinforcement learning.

AI Expert Training Course (KAIST IT Academy)

Sep 2024 – Nov 2024

- Completed rigorous certification focusing on big data analysis, statistical mathematics, and ML/DL theories.
- Gained hands-on proficiency in deep neural networks, multilayer perceptrons, data preprocessing, and optimizer tuning.

SELECTED ACADEMIC COURSEWORK

- Introduction to Artificial Intelligence (Honors)
- Advanced Data Structures & Algorithms
- Discrete Mathematics & Computer Systems
- Linear Algebra & Applied Probability

TECHNICAL SKILLS & CERTIFICATIONS

- **Programming Languages:** Java, Python, PyTorch, C.
- **Certifications:** Professional Level Specialist Certificate in Java.